Factor Risk Model — Response to the Risk Review

Mapping Chris's notes to what's in place, what we've built, what's designed, and what we propose — with methodology where it matters.

This is a proof-of-concept Barra-style equity factor-risk model, built on free/public data and demoed against the Soros 13F book. Chris's notes from the walkthrough are the agenda below. Each point is classified by status and answered with the methodology, not just an intention. The honest framing he gave us — *the devil is in the details* — is the one we're holding ourselves to: quality lives in exactly how loadings are built, how the universe is picked, how data is standardised, and how missing data is handled.

● **In place** — already true today ● **Done** — built in response to this review

● **Designed** — specced, awaiting sign-off ● **Proposed** — on the roadmap

1 Point-in-time data ● IN PLACE

> **Chris:** "All the data needs to be point in time to avoid biases in the estimation of the model."

This is already a core principle, not an add-on. Nothing enters a date's estimation that wasn't public by that date:

- **Fundamentals** are as-of joined on the SEC `filed` date (a backward merge), so a metric is only visible from the day it was reported — never back-dated to the period it describes.
- **13F positions** are as-of joined by *filing* date, lagged the same way; exited names expire on the next filing rather than persisting.
- The calendar is monthly month-ends, and the universe is rebuilt each month (see §2).

No change required. We flag it because it directly answers the bias concern.

2 Estimation vs. coverage universe ● DESIGNED

> **Chris:** "Estimation universe needs high-quality data … Coverage universe needs to contain all names on which you have (or may want) positions. The universes change monthly. Choosing the universe has an impact on the results — e.g. Russell 3000 vs. SPX; simplest example: 0-filled missing values."

Today: there is one universe. It is the 13F-held names unioned with an index seed (SPX), used *both* to estimate factor returns *and* to carry positions. The 0-fill he mentions is already how thin names are handled — a name with at least 6 of 10 loadings is kept and the rest are zero-filled (the market-average tilt).

Proposed: split the one universe into two, both rebuilt monthly.

UNIVERSE	MEMBERSHIP	ROLE	LOADINGS CAPPED?
Estimation	high-quality liquid names (index seed + a data-quality gate)	regress factor returns here only	Yes — winsorised
Coverage	estimation ∪ every held (and candidate) name	assign loadings to all of these	**No**

Full spec, builder changes, and the four open questions for Chris are in the companion design note (`docs/estimation-coverage-design.md`). Nothing downstream changes — the six data frames, the cube, and every risk measure are untouched; this only changes *how* loadings and factor returns are produced.

OPEN DECISION FOR CHRIS — ESTIMATION UNIVERSE BREADTH

Proposal: estimate on the **S&P 1500**; let **coverage** carry Russell-3000-level breadth as the book requires. Today we estimate on the S&P 500 seed (plus held names), which is a touch narrow for a stable daily cross-sectional regression.

Why SP1500, not raw Russell 3000, for estimation. The two universes want opposite things. Estimation wants breadth *and* clean data — your "high-quality data" point. Russell 3000 reaches deep into micro-caps where our free sources degrade (sparse/late SEC fundamentals, illiquid prices, the corrupt-market-cap cases we already guard against), and that noise pulls *every* factor return. SP1500 roughly triples the cross-section while staying liquid and well-covered. The genuine case for R3000 breadth is **coverage**, which must contain every held name anyway — and §3's uncapped coverage loadings already let a small off-index name read its true large-negative Size loading.

Two caveats to weigh. (1) *Point-in-time index membership* — we currently apply today's index list across all history (a survivorship/look-ahead bias, the very thing §1 avoids elsewhere); broadening makes a clean historical-membership source harder, especially for R3000. (2) *Pull scale* — roughly 3× the per-name data pull for SP1500, ~6× for R3000.

How we'd settle it: an empirical sweep — SP500 vs SP1500 vs a liquidity-screened R3000 — compared on factor-return stability (vol, regression condition number), cross-section breadth per day and the `<30 valid names` skip rate, and the Kupiec/Basel backtest. Pick the universe where factor returns are most stable and the backtest stays green — the same evidence-led approach we used to choose the FHS backtest estimator.

For Chris: is SP1500 the right estimation set, or do you want a screened Russell 3000? Any required liquidity / market-cap screen, and is a point-in-time membership source available to us?

3 Cap the estimation universe, not the coverage universe ● DONE — SHIPPED

> **Chris:** "If your estimation universe is SPX and your coverage universe is Russell 3000, the small names will have very negative size loadings, and that is correct."

This is now exactly his last point. Previously the standardiser `_winsor_z` clipped every loading at ±3σ for every name — estimation and coverage alike — so a genuinely tiny holding's Size loading was clipped to −3 instead of reading its true, larger-magnitude value. The standardiser is now `_split_z`, which keys off the §2 estimation flag and clips only estimation names.

Methodology — the standardisation

A descriptor `x` is turned into a loading by a robust z-score against the cross-section's median and MAD (median absolute deviation, scaled to a normal σ):

```
med   = median(x)
MAD   = 1.4826 · median(|x − med|)
z     = (x − med) / MAD
load  = (z − mean(z)) / std(z)        ← re-standardised to mean 0, std 1
```

`_split_z` splits this in two, keyed off §2:

- **Estimation names:** compute med/MAD here, and *winsorise* — clip `z` to ±3 — before regressing. This protects the regression from a corrupt or extreme descriptor blowing up every factor return (the historical condition-number failures).
- **Coverage names:** standardise against the *estimation* universe's med/MAD, and leave the loading **uncapped** — only a loose ±10 backstop (`COVERAGE_CAP`) clamps a corrupt XBRL value. A held name far smaller than anything in SPX correctly reads, say, −6 — six estimation-MADs below the median. That is a real exposure; capping it would understate the book's size tilt.

Capping belongs in estimation (clean fit), not in coverage (faithful exposure). Exactly his point.

4 Drawdown ● DONE — SHIPPED

> **Chris:** "Drawdown is an important lens that's currently missing."

Built and live. VaR and Expected Shortfall are single-horizon distribution statistics — they say nothing about the *path*. Drawdown is the path lens: the worst peak-to-trough an investor would have sat through.

Methodology — constant-portfolio drawdown

We already simulate the current book's daily P&L across the full factor-return history (the scenario engine's P&L vector, date-ordered). Drawdown compounds that into an equity curve and reads the deepest trough below a running peak:

```
equity_t   = ∏ (1 + pnl_s),  s ≤ t      ← held book, compounded over the path
peak_t     = max(equity_s,   s ≤ t)
drawdown_t = equity_t / peak_t − 1       ← ≤ 0
max DD     = min_t drawdown_t
```

It also reports the peak and trough dates, whether and when it recovered, and the longest run spent underwater. Like the VaR backtest, this is a *constant-portfolio what-if* — today's book applied to history — not a live P&L track record (the 13F book has none).

What it shows on the Soros book (HistFull, as of 2024-12-31)

MEASURE	VALUE
Max drawdown	−38.96%
Peak → trough	2020-02-19 → 2020-03-23
Recovered by	2020-06-08
Longest underwater	524 trading days
vs. worst single day (1-day)	−13.2%

The lens earns its place immediately: the static 99% daily VaR reads a comfortable number, but the COVID path takes this book down ~39% and keeps it underwater for two years. That drawdown comes through Market beta the one-day VaR cell underprices. It surfaces three ways — a `/drawdown` API endpoint, an equity-curve + underwater panel in the dashboard, and folded into the written risk commentary so it leads with a deep drawdown when present.

5 Active risk & active return vs. a benchmark ● AGREED — NEXT

> **Chris:** "'Active risk' and 'active return', where you measure your performance against a benchmark portfolio, can be more important than stand-alone risk/return."

Today the model is entirely absolute — stand-alone risk of the Soros book, with no active view yet. **The benchmark is now agreed: MSCI Large Cap**, so this moves from an open question to the next build.

Methodology — what we'd add

Once the coverage universe (§2) is explicit, a benchmark is just another coverage portfolio. Active risk falls straight out of the same factor machinery on the *active weights*:

```
w_active = w_portfolio − w_benchmark
x_active,k = Σ_i w_active,i · loading_i,k     ← active factor exposures
tracking error² = x_activeᵀ · F · x_active  +  Σ_i w_active,i² ·
σ²_spec,i
```

That gives tracking error, active VaR/ES, and a by-factor decomposition of active risk — the same marginal/incremental contribution views we already compute for absolute risk, but relative to the benchmark. **Depends on §2** — the MSCI Large Cap constituents must sit in the coverage universe. Pairs naturally with §6.

6 Intentional vs. unwanted loadings — hedging ● PROPOSED

> **Chris:** "Some factor loadings are wanted/intentional positioning, some are unwanted/accidental and can/should be hedged out." Also: "Market/factor risk can be hedged at the portfolio level or at the firm level."

We report exposures and per-factor marginal/incremental risk, but nothing yet distinguishes a deliberate tilt from an incidental one, and there's no hedge view.

Methodology — what we'd add

- Tag each factor (or active exposure) as **intended** or **incidental** in config.
- Split active risk (§5) into the part explained by intended bets vs. the residual incidental factor risk — the hedgeable slice.
- Report the hedge that neutralises the incidental exposures (the offsetting factor positions), at the portfolio level, with a note on where firm-level netting would change it.

This is mostly a reporting/decomposition layer on top of §5, not new estimation.

7 Orthogonalisation & multicollinearity ● PARTIAL → PROPOSED

> **Chris:** "You can orthogonalize factors to avoid any issues with multicollinearity."

Today, partially: we orthogonalise NonLinSize against Size (regress out the linear part, then re-standardise), and a degeneracy screen drops any factor whose cross-sectional spread collapses on a given day, which keeps the regression conditioned. We do not yet orthogonalise the full factor set.

Methodology — what we'd add

- Full factor orthogonalisation (a defined order, or a symmetric scheme) so correlated descriptors don't split a single bet across two coefficients.
- Surface the diagnostics that justify it: pairwise factor correlations, VIF per factor, and the design-matrix condition number per date — so the choice is evidenced, not asserted.

8 Linked securities ● PROPOSED

> **Chris:** "Linked securities break the assumption that the idiosyncratic returns are uncorrelated (different share classes, dual listings, ADRs and underlying, etc.). There are ways to handle that."

Today this assumption is explicit and unhandled: the specific-risk block is strictly diagonal — idiosyncratic returns assumed uncorrelated. Linked names (GOOGL/GOOG, an ADR and its local line, dual listings) violate it: their residuals move together, so the diagonal understates their joint specific risk and a concentration measure treats them as two independent bets when they are one.

Methodology — what we'd add

- Identify linked groups (share-class / ADR / dual-listing maps).
- Either **net** linked names into a single economic position before the specific block, or add **off-diagonal specific covariance** within each linked group — a block-diagonal specific matrix rather than fully diagonal.

This touches the model's core "diagonal specific block, no full specific covariance" assumption, so it needs a short design pass before code — flagged, not hidden.

9 PnL attribution — realised PnL by factor and residual ●

DESIGNED

> **Chris:** "Show how the factor model explains the realised PnL over a period — broken down by factor and by residual — and check whether the residuals are large, and if they are, whether they're correlated; ideally uncorrelated, which shows the PM is doing a good job."

> **Chris, 29 Jun (endorsed):** "It's really the connection between the PnL attribution and the risk decomposition at the start of the period that tells you how good the risk management of the portfolio is."

A performance-attribution report: split the book's realised PnL into what the factor model explains and what it doesn't, then test the residual.

What Chris asked for — the full picture

The request is not a PnL attribution on its own. It's a **loop between two reports**, and the value is in connecting them.

- **Ex-ante, at the start of the period (date T) — a risk decomposition.** Which factors and which positions contribute how much to the book's **total risk**. You don't forecast

returns, but from each factor's volatility and the pairwise correlations you get the *shape and width* of the forward PnL distribution (its VaR / ES); shocking those vols and correlations — or replaying a past episode — gives stress scenarios. What you forecast is the *range of outcomes*, never a number.
- **Ex-post, over the period (T→T+1) — a PnL attribution.** Realised PnL split into the same factor and residual buckets.
- **The connection.** Was the realised PnL in the range the risk decomposition implied, and did the contributions line up with the risk actually taken? In his words: *"it's really the connection between the PnL attribution and the risk decomposition at the start of the period that tells you how good the risk management of the portfolio is."*

His day-to-day loop:

1. Run the risk decomposition at the start of the period.
2. Check the exposures are within limits, are the intended bets, and are in line with the mandate.
3. If not — exit the offending positions, or add hedges.
4. Run the PnL attribution at the end, and check the explanation matches what the risk decomposition led you to expect.

Two checks close the loop. **Within band** — does each factor's realised contribution sit inside the band its start-of-period risk implied? This is *not* "big-risk factors should drive the PnL": if the big-risk factor doesn't move and a small-risk one falls off a cliff, the small one rightly dominates — what matters is each contribution against its *own* risk-implied range, which already folds in the move. **The surprise** — a realised contribution outside that band, an outcome the risk decomposition didn't see coming.

	PNL: MADE MONEY	PNL: LOST MONEY
Risk you knew you had	bet paid — fine, you chose it; some bets win	bet lost — fine, you chose it; some bets lose
Risk the decomposition missed	**investigate** — a win you can't explain	**investigate** — a loss you can't explain

Chris's correction: an *unexpected gain is as much of a risk-process failure as an unexpected loss* — "in one case the investors are happy, in the other angry, but both show the same failure." The axis that matters is **known vs. unknown risk**, not made vs. lost.

The principle behind it, in his words: **the primary function of the risk team is not to avoid losses but to understand all the risks the fund is taking.** Some bets pay, some don't — that's the business, and not on the risk manager. But making *or* losing money on a bet you didn't know you took is fully on the risk manager: the direction was luck, the failure was not seeing the risk. So good risk management is measured by **how well the PnL is explained by the risk you knew you had**, not by the sign of the PnL. (A secondary function is to keep those bets from threatening the fund — limits and mandates, capital and liquidity reserves, hedging.)

He frames it as a universal pattern, with two other examples. In the **sensitivities** world: the Greeks report at the start vs. the PnL-explain over the period — if they don't line up, there are second-order or non-linear effects the sensitivities missed. In the **VaR** world: the realised PnL distribution vs. the VaR — too many exceptions means the VaR model is missing something (our `/backtest` is exactly this). Same loop every time: form an ex-ante expectation, measure ex-post reality, and treat the unexplained part as the signal.

Get one thing straight first. A factor risk model explains return after the fact — it does not forecast it. The "factor PnL" below is not a prediction of anything; it is the part of the *realised* return the factors explain. (We avoid the word "projected" deliberately, per Chris — there is no forward PnL forecast.)

```
R_p(t) = Σ_k x_k(t)·f_k(t)  +  u_p(t)
x_k = Σ_i w_i·loading_i,k     ← book net exposure to factor k
u_p = Σ_i w_i·ε_i             ← residual (specific) return
ε_i = R_i − Σ_k loading_i,k·f_k   ← the WLS residual the fit already
computes
```

The factor PnL is `Σ_k x_k·f_k`; the residual is `u_p`. They sum to realised by construction, because `ε_i` is the same residual `regress_factors` already produces and currently discards (only its square is kept, as `SpecificVar`). We keep that residual as a new `specific_returns` frame so the cube can carry it as a measure — it is the model's own number, not an approximation.

Methodology — the realised engine

Realised PnL is built bottom-up from the ~1,000 name price series, *not* from the cube (which has no realised-PnL measure, only hypothetical scenarios):

- **Unit NAV per name** — daily close → daily return → cumulative product, an index base 1.0.
- **Drifting (buy-and-hold) weights** — anchor at each 13F filing, then let each name's NAV compound so the book weights drift with the market until the next filing. That

is what actually happened to the book between filings; it is more honest than the constant-portfolio assumption the drawdown and backtest lenses use.
- **Price-only**, both sides — the factor model is price-only, so the realised return fed in is the same return the regression saw. Mixing in a dividend-adjusted series would make the residual absorb every dividend as fake alpha. Dividends are excluded both sides, disclosed.

The decomposition then ties out three ways — realised (bottom-up) = Σ factor contribution + specific, to machine precision daily — and is Carino-linked over the period so the by-factor contributions sum exactly to the compounded book return with no plug.

Methodology — the residual diagnostics

The residual is the part the model can't explain. For a name, `R_i = Σ_k loading·f_k + ε_i`; `ε_i` is the **specific (idiosyncratic) return**, the stock-specific move the factors miss, and at the book level `u_p = Σ_i w_i·ε_i`. The premise: if the residual is genuine, diversified stock-picking, the specific returns behave like independent coin-flips — no pattern over time, no link to the factors, no common thread across names, sized about as the model predicted. Each test checks one of those.

Is it large?

- **Specific share** — cumulative specific PnL as a fraction of total: how much of the return came from stock-picking vs. factor bets.
- **Information ratio** — `mean(u_p)/std(u_p)`, annualised: a Sharpe ratio for the idiosyncratic slice, the stock-picking skill number. Positive and high = reliable alpha; near zero = noise; negative = stock-picking is destroying value.
- **Realised vs. predicted specific vol** — realised `std(u_p)` against the model's predicted specific vol. Ratio ≈ 1 = sized right; above 1 = the model under-states idiosyncratic risk, which makes total VaR too low.
- **Explained share** — `1 − var(u_p)/var(R_p)`, the fraction of the book's return the factors explain. Low isn't automatically bad — diversified stock-picking *or* a missing factor; the next three tests tell them apart.

Is it correlated? — three places a hidden pattern can hide. Chris's steer: keep these cheap — the simple correlation and regression versions are enough, and the formal tests are noted but optional, since the desk quants already know them.

- **Across time.** Does this month's specific return predict next month's? Correlate `u_p` with itself lagged 1 and 2 periods. A non-zero correlation (a persistent trend) means the “alpha” is a slow, unhedged systematic bet, not nimble stock-picking. (The formal version is the Ljung-Box white-noise test across many lags — optional; the lag-1/2 correlation is the cheap read.)

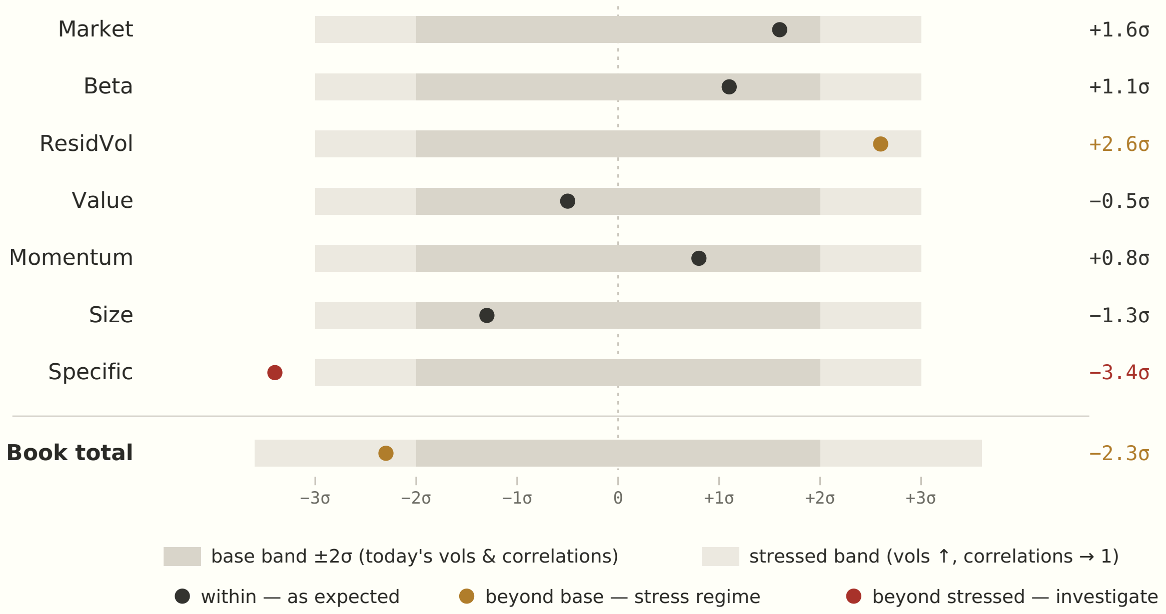
- **Against the factors.** Regress the specific return on the factor returns. It is orthogonal within one month by construction, but over time — and because the 13F exposures are lagged up to 45 days — it can still co-move with a factor. A loading means part of the “alpha” is that factor's beta mis-measured: paid for systematic risk and calling it skill. Near zero = clean, orthogonal alpha. (Chris flagged this regression as one of the worthwhile tests to build.)
- **Across names.** The model *assumes* the names' specific returns are uncorrelated (the diagonal specific block). If many names' residuals move together, that is **most likely a missing factor** — a common driver the model has no factor for; a small linked group (share classes, ADRs) is the narrow special case. A shared residual means the diagonal block **under-states risk** — what looks like many independent bets is one bet repeated. (The formal version is a PCA on the residual panel — optional / deferred per Chris, unless quick.)

Bias statistics — did the model size the risk right? The canonical Barra / MSCI check. Standardise each realised return by its *predicted* vol, `z = realised / predicted-vol`; if the forecast is right, `z` has standard deviation ≈ 1. The bias statistic `B = std(z)` over a rolling window reads: `B ≈ 1` calibrated, `B > 1` risk under-forecast (the dangerous direction), `B < 1` over-forecast. Run it for the whole book, each factor, and the specific block.

The read for Chris: positive IR, no lag-1/2 autocorrelation, ~zero factor correlation, no common residual component across names, and bias ≈ 1 → genuine, diversified, well-measured stock-picking, the PM is doing well. Autocorrelation (a persistent unhedged bet), factor correlation (hidden beta), or a dominant residual component (a missing / crowded factor) each name a specific problem → tell the PM, add the factor, or hedge it. Also surfaced: by-sector / by-name attribution, residual concentration (HHI), hit rate, and factor-timing vs. static tilt.

Reconcile view — realised vs. the expected band

The headline of the linkage is one figure. Each factor's **realised contribution** over the period (the dot) is read against the **band the risk decomposition implied at the start** — centred at zero (the model forecasts the *width* of the distribution, not its direction), with half-width equal to that factor's σ. Two bands per row: the **base** forecast, and a **stressed** one with vols shocked up and correlations pushed toward 1. The band is just the covariance applied to the exposures ($\sigma^2 = x^{\top} \cdot F \cdot x$), so the stressed band *is* Chris's "shock the vols and correlations", drawn.



Illustrative — a stressed quarter on the Soros book. Each row is read against its own start-of-period band; the dot is the realised contribution. **Market** and the style factors land inside their bands — risk understood, PnL in proportion to it. **ResidVol** and the **book total** breach the base band but sit inside the stressed one: a stress regime, where the calm-market estimate was too benign but the shocked one covers it. The book's band widens more than any single factor's because **correlations → 1 removes the diversification** that narrowed it — the aggregate is where correlation risk surfaces. **Specific** is the flag: it lost more than even the stressed band predicted — the "lost money where the risk decomposition said little" case, a risk not captured at the start (here a crowded / linked bet the diagonal specific block understates), the thing to dig into.

Drill-through — parent to child

The factor breakdown is the parent; clicking a factor expands to the names beneath it — *which positions made that factor bet* — and the column still sums to the factor row above. It foots because the parent value is itself a sum over names ($x_k \cdot f_k = \Sigma_i w_i \cdot \mathrm{loading}_{i,k} \cdot f_k$), and the same data drills both ways: factor → name, or name → factor (why a position made money, its factor mix plus specific). This is built **cube-native**: three additive measures (`Factor contribution` , `Specific PnL` , `Realised PnL`) in the existing pivot grid, so the drill, the slicing and the re-pivot are native — and the written commentary and Q&A inherit them.

This is the general principle, not just for attribution: **additional measures live in the cube** — including the more complex ones the quants define — whenever it makes sense to apply them to data we **slice, drill, or modify**. A measure defined once in the cube composes with everything: any slice (factor, sector, name, date, scenario), the parent→child drill, and — crucially — recomputation under **what-if** (modified weights, added or dropped names) and **stress** (shocked vols / correlations). So a quant's measure is immediately available across every view and every hypothetical, rather than being re-coded for each context. Only the genuinely non-cube statistics (period linking, the residual correlation tests, bias) stay in the thin Python layer.

What we already have, and what we'd build

Most of the ex-ante half already exists in the model — it's the connection and the attribution measures that are new.

PART OF THE REQUEST	ALREADY IN PLACE	TO BUILD
PnL attribution by factor + residual (ex-post, T→T+1)	factor returns, exposures, positions, prices; the decomposition identity `R_p = Σ x_k·f_k + u_p`	the realised engine (unit NAVs, drifting weights); cube measures `Factor contribution` / `Specific PnL` / `Realised PnL`; the `specific_returns` frame; `/pnl_attribution` + panel
"Are the residuals large / correlated?"	predicted specific vol (`SpecificVar`); the residual itself (the model's own number)	information ratio + the var-ratio measures; the residual-vs-factor regression; lag-1/2 autocorrelation; bias statistics. Ljung-Box / PCA optional (the quants know them); `/pnl_attribution/residual`
Risk decomposition at T — factor & position contribution to total risk	**the cube already computes this** — marginal / incremental contribution to Total VaR; the Risk-HHI concentration measure is built from these shares	surface it as a tidy "contribution to risk at T" table aligned to the same factor / position buckets as the attribution
Expected forward distribution — vols + correlations, stress / replay	scenario VaR / ES over full history (empirical), event-replay sets, hypothetical shocks, `/stress` (per-factor vol shocks)	the ±σ **base band** per row; a **correlation-stress mode** for the stressed band — today `/stress` shocks vols only, not correlations

PART OF THE REQUEST	ALREADY IN PLACE	TO BUILD
The linkage / reconcile — risk at T vs. PnL T→T+1, within-range, surprise	`/backtest` (realised-vs-VaR exceptions, Kupiec / Basel — his VaR analogy); the what-if risk math (book risk recomputed from weights)	the temporal pairing; the per-factor / position surprise z-score; the surprise ranking; the reconcile band chart above; `/pnl_attribution/linkage`
The daily loop — decompose → check limits / intended → hedge → attribute	`/limits` (the limits check); style-drift & what-changed (intended vs. unintended); what-if & stress (the hedge step)	no new engine — the linkage report is what closes the loop end to end

The cube measures are additive (the same class as net exposure — no heavy scenario vectors, so no performance penalty). Persisting the residual adds a seventh data frame (`specific_returns`), a deliberate, contained change that needs a rebuild — the price of a live reconciling drill rather than a static table. The statistics the cube can't express (period linking, the residual correlation tests, bias stats) sit in a thin Python layer: `barra_pnl_attribution.py` , the `/pnl_attribution` + `/pnl_attribution/residual` endpoints, and a dashboard panel. A by-product is a true realised track record, which upgrades the constant-portfolio inputs the drawdown and backtest lenses use today. Full spec in the companion note `docs/pnl-attribution-plan.md` .

Roadmap at a glance

#	ITEM	STATUS	EFFORT	UNLOCKS / NOTE
1	Point-in-time data	● In place	—	bias-free estimation
4	Drawdown lens	● Done	shipped	path risk VaR/ES miss
2 · 3	Estimation/coverage split + uncapped coverage	● Done	shipped	correct off-index loadings; clean fit; foundation for §5–6
5	Active risk vs. benchmark	● Agreed — next	project	benchmark agreed: MSCI Large Cap
6	Intentional vs. unwanted (hedging)	● Partial	evidence shipped	style-drift attribution panel: rotation vs loading drift; final call needs desk intent

#	ITEM	STATUS	EFFORT	UNLOCKS / NOTE
7	Full orthogonalisation + diagnostics	● Partial	moderate	builder change
8	Linked securities	● Proposed	design first	block-diagonal specific risk
9	PnL attribution & factor-model validation	● Designed — endorsed	~1.5–2 wk	realised PnL by factor + residual; residual correlation tests; linked to the risk decomposition at T

Recommended order: **drawdown** (done) → **estimation/coverage split** (done — the keystone, now live and the basis for the span and style-drift panels; it unblocked the uncapped coverage loadings) → the **intentional/unwanted** evidence is shipped as the style-drift attribution panel. **PnL attribution** (§9) is specced and endorsed — the natural next build → then **active risk**, whose benchmark is now agreed (MSCI Large Cap), with orthogonalisation diagnostics and linked securities as they fit.

Generated for the factor-model risk review. Methodology cross-refers to `barra_build_frames.py` *(estimation),* `barra_factor_risk_cube.py` *(risk engine),* `risk_api.py` *(measures & drawdown), and the design note* `docs/estimation-coverage-design.md`*.*